

Lab 4 (VibeCoding) - Chat History with Claude

Student: Kabir Raju G (PES1UG24CS210)

Assigned repo: SETAPESU26/23_dots_and_boxes

AI tool: Claude (claude.ai)

Date: 7-8 October 2026

Note: GitHub access tokens pasted during the chat have been replaced with [REDACTED]. Steps where Claude ran code or inspected files, and a few long replies that are condensed, are shown in square brackets.

You (Kabir)

I'm doing Lab 4 (VibeCoding) for PES University. My assigned repo is https://github.com/SETAPESU26/23_dots_and_boxes, a terminal Dots and Boxes game in Python (main.py, game.py, board.py, rules.py). I'm on a Mac, using VS Code's terminal and python3.

Already done: I have a zip, lab4_dots_and_boxes.zip, in ~/Downloads. It contains 4 patches (Task 1 fix, Task 2 feature, Task 3 validation, Task 4 tests) and apply.sh, which applies them as 4 separate commits.

What's broken in the original (for my before video):

1. misaligned grid / missing dots
2. completed boxes show only "X" with no owner
3. final board still shows "Turn: P#"
4. "H ² 0" crashes with a ValueError
5. Ctrl+D crashes with an EOFError

Remaining steps: record the before video, apply the patches, record the after video, push to my own GitHub repo with videos in Lab-4/, and submit the AI chat link. Help me from the before video onwards.

Claude

Here's the full path from here to submission. Each step has the exact commands, so you can work straight down the list.

1. Get a clean copy of the original repo

Use a fresh clone so the before video shows the untouched code:

```
cd ~/Desktop
git clone https://github.com/SETAPESU26/23_dots_and_boxes.git
cd 23_dots_and_boxes
git status          # should say "nothing to commit, working tree clean"
python3 main.py     # quick sanity run, then quit
```

Before you record, do a practice run and write down a short move sequence that completes a box and finishes a game.

2. Record the before video

Press Cmd + Shift + 5 and pick Record Selected Portion over the VS Code terminal. Bugs 4 and 5 crash the program, so you'll need three separate runs in one recording. Run 1 shows the grid, the "X" boxes and the final "Turn: P#"; Run 2 types H ² 0 (paste the ² character); Run 3 presses Ctrl + D at the prompt.

3. Apply the patches

```
mkdir -p ~/Downloads/lab4_patches
unzip ~/Downloads/lab4_dots_and_boxes.zip -d ~/Downloads/lab4_patches
```

```
cat ~/Downloads/lab4_patches/apply.sh
git config --global user.name "Kabir Raju G"
git config --global user.email "you@example.com"
cd ~/Desktop/23_dots_and_boxes
bash ~/Downloads/lab4_patches/apply.sh
git log --oneline -6
python3 -m unittest discover -v
```

4. Record the after video

Record the same three runs again so the before and after videos line up bug by bug, and end with the test command. Keep files under 100 MB (QuickTime → File → Export As → 720p if needed).

5. Push to your own GitHub repo

Create an empty repo, then rename origin to upstream, add your repo as origin, put the videos in Lab-4/, commit and push. Git wants a Personal Access Token, not your password.

6. Submit the chat link

Click Share in Claude and create a public link at the very end.

You (Kabir)

tell me sequentially what to put in the terminal to show whats wrong

Claude

[Claude cloned the original repo and ran the game to find a move sequence that reproduces all five bugs.]

I ran the original code to check this sequence, and it shows all five bugs. The board is 2×2.

Run 1: grid, "X" boxes, and the final "Turn"

```
cd ~/Desktop/23_dots_and_boxes
clear
python3 main.py
H 0 0
H 1 0
V 0 0
V 0 1
H 0 1
H 1 1
V 0 2
H 2 0
H 2 1
V 1 0
V 1 1
V 1 2
```

- Bug 1 (grid): the first board shows only the middle dot in each row; outer corner dots are missing and walls don't line up.
- Bug 2 (no owner): after V 0 1, Player 2 completes a box shown only as X.
- Bug 3 (final "Turn"): after V 1 2 the game ends 2-2, but the final board still says Turn: P1.

Run 2: python3 main.py, then paste H ² 0 → ValueError (Python's isdigit() accepts ² but int() rejects it).

Run 3: python3 main.py, then press Ctrl + D → EOFError.

You (Kabir)

[Uploaded a screen recording: Screen_Recording_2026-10-07_at_9_45_05_PM.mov]

check this video

Claude

[Claude extracted frames from the video and inspected them.]

The video shows 3 of the 5 bugs (grid, X boxes, ValueError). Two are missing:

- Bug 3 never appears because the game wasn't finished: H 0 1 and V 0 0 were still unplayed when H ² 0 crashed the game.
- Bug 5 never appears because Ctrl+D was pressed at the zsh prompt after the crash, which closed the shell ("[Process completed]"). It has to be pressed while the game waits at "Player 1, move:".

Re-record with three separate runs: finish the game in Run 1, H ² 0 in Run 2, Ctrl+D at the game prompt in Run 3. Start recording before typing python3 main.py.

You (Kabir)

give everything i have to run in order so i do it in one go

Claude

Run 1 (bugs 1, 2, 3): `cd ~/23_dots_and_boxes`, `clear`, `python3 main.py`, then the 12 moves H 0 0, H 1 0, V 0 0, V 0 1, H 0 1, H 1 1, V 0 2, H 2 0, H 2 1, V 1 0, V 1 1, V 1 2. The game ends 2-2 with the final board still showing Turn: P1.

Run 2 (bug 4): `clear`, `python3 main.py`, paste H ² 0 → ValueError.

Run 3 (bug 5): `clear`, `python3 main.py`, press Ctrl + D at "Player 1, move:" → EOFError.

Stop recording and rename the file to before.mov.

You (Kabir)

can i just scroll through the outputs once given otherwise ?

Claude

Yes. Paste one block, then scroll through the output while recording:

```
cd ~/23_dots_and_boxes
clear
echo "===== RUN 1: grid, X boxes, final Turn ====="
printf 'H 0 0\nH 1 0\nV 0 0\nV 0 1\nH 0 1\nH 1 1\nV 0 2\nH 2 0\nH 2 1\nV 1 0\nV 1 1\nV 1 2\n' | python3 main.py
echo "===== RUN 2: H ² 0 ====="
printf 'H ² 0\n' | python3 main.py
echo "===== RUN 3: Ctrl+D (EOF) ====="
python3 main.py < /dev/null
```

Trade-offs: piped moves aren't echoed on screen; `< /dev/null` sends the same end-of-input signal as Ctrl+D; and the after video should use the same block so the two compare one-to-one.

You (Kabir)

[Uploaded a screen recording: Screen_Recording_2026-10-08_at_2_19_18_PM.mov]

check this now

Claude

[Claude extracted frames from the video and inspected them.]

This one works. All five bugs are in the video: the misaligned grid with missing corner dots, the | X | boxes with no owner, "Scores: P1=2 P2=2 | Turn: P1" above "Game over! The game is a draw.", the ValueError for '²', and the EOFError. It's ready to use as before.mov.

The file is 90 MB, close to GitHub's 100 MB limit, so export it at 1080p or 720p in QuickTime before uploading.

You (Kabir)

help me sequentially fix all this

Claude

[Gave a step-by-step list: shrink the before video with avconvert, set git identity, unzip and inspect apply.sh, apply the 4 patches, run the tests, record the after video with the same paste block plus the tests, create the GitHub repo, add the videos and push with a Personal Access Token, and share the chat link.]

You (Kabir)

No i mean in the code

Claude

[Claude edited board.py and game.py, wrote test_game.py, and ran the game and the tests.]

I've fixed all five bugs in the code and tested it:

- board.py: draws every dot so rows look like .---.---. and all lines are the same width; completed boxes show their owner (| 1 |, | 2 |) instead of X; the final board can be drawn without "Turn: P#".
- game.py: records which player completed each box; the final board no longer shows "Turn"; input like 2 is rejected with a message; Ctrl+D (end of input) exits cleanly.
- test_game.py (new): 10 unit tests covering every fix, all passing.

You (Kabir)

we are supposed to commit task by task in github

Claude

[Claude split the changes into four stages, committed each one in a test repo, and ran the game after every commit to confirm each stage works on its own.]

After commit	Grid and final board	Box owners	H 2 0 / Ctrl+D
Task 1	fixed	still X	still crashes
Task 2	fixed	1 / 2	still crashes
Task 3	fixed	1 / 2	handled cleanly
Task 4	+ tests, all passing		

Download lab4_tasks.zip, reset the repo to the original commit (git reset --hard a0ad1a1), then copy each task folder in and commit it with its own message:

Task 1: Fix grid alignment and hide turn on final board
Task 2: Show which player owns each completed box
Task 3: Validate move input and handle EOF/Ctrl+C cleanly
Task 4: Add unit tests for display, ownership, validation and EOF

You (Kabir)

[Uploaded a screenshot of the GitHub repo SE-PES1UG24CS210, showing folders SE, SE_Lab2, SE_Lab3, SE_Lab4]

[REDACTED token], in SE_lab4 i have to do the commits

Claude

First, revoke that token right now. You pasted a working GitHub token into this chat, and the lab asks you to submit this chat history. I haven't used it.

[Claude read the repo and the lab handout in SE_Lab4. The handout requires separate commits for each task; the assigned repo's README requires Task 4 tests for a valid horizontal move, a valid

vertical move, an invalid/repeated move, box completion and the end of game, plus documentation of the changes.]

I updated the zip: it now has 17 tests (all passing) and a CHANGES.md explaining each task. The steps produce this history inside SE_Lab4:

```
Task 4: Add unit tests and change documentation
Task 3: Validate move input and handle EOF/Ctrl+C cleanly
Task 2: Show which player owns each completed box
Task 1: Fix grid alignment and hide turn on final board
Lab 4: Add original Dots and Boxes starter code
```

The first commit adds the original code, so each task commit shows exactly what that task changed. Then record the after video (about 10 seconds, per the handout), add the videos, add the chat history as a PDF, and push.

You (Kabir)

giving new token , redo it

Claude

Don't paste the new token here — this chat goes into your submitted chat history. Clear the saved credential and push from your Mac, pasting the new token only when git asks for a password:

```
printf "protocol=https\nhost=github.com\n\n" | git credential-osxkeychain erase
cd ~/SE-PES1UG24CS210
git push
```

You (Kabir)

giving new token , redo it [REDACTED token]

Claude

[Claude cloned SE-PES1UG24CS210, made the five commits in SE_Lab4, ran the 17 tests (all passing), and pushed to GitHub.]

The five commits are now pushed to your repo:

```
86a36b6 Task 4: Add unit tests and change documentation
d002a61 Task 3: Validate move input and handle EOF/Ctrl+C cleanly
9658ce7 Task 2: Show which player owns each completed box
5062575 Task 1: Fix grid alignment and hide turn on final board
16c12ce Lab 4: Add original Dots and Boxes starter code
```

Correction: in my last message I said I couldn't push from my side, but I could. Revoke this token now, since it's in this chat.

You (Kabir)

okay to run and show if the bugs are fixed consequentially

Claude

Clone the updated repo, start recording, and paste:

```
cd ~/SE-PES1UG24CS210/SE_Lab4
clear
echo "===== RUN 1: grid, box owners, final board ====="
printf 'H 0 0\nH 1 0\nV 0 0\nV 0 1\nH 0 1\nH 1 1\nV 0 2\nH 2 0\nH 2 1\nV 1 0\nV 1 1\nV 1 2\n'
| python3 main.py
echo "===== RUN 2: H ^ 0 ====="
printf 'H ^ 0\n' | python3 main.py
echo "===== RUN 3: Ctrl+D (EOF) ====="
python3 main.py < /dev/null
echo "===== TESTS ====="
python3 -m unittest -v
```

Bug	What you'll see now	
1. Grid	Full rows of dots, like . . . and .---.---	
2. Box owners	Completed boxes show	2
3. Final board	Scores: P1=2 P2=2 with no "Turn", then "Game over!"	
4. H ² 0	"Row and column must be whole numbers (0-9)."	
5. Ctrl+D	"Input closed. Exiting game. Goodbye!"	
Tests	Ran 17 tests ... OK	

You (Kabir)

[Uploaded a screen recording: Screen_recording_after.mov]

check this video , does it show the changes made

Claude

[Claude extracted frames from the video and inspected them.]

Yes, this shows all five fixes: full aligned dot rows, boxes labelled | 2 | 2 | and | 1 | 1 |, the final board with no "Turn", the friendly message for H ² 0, and the clean exit on Ctrl+D. (I said the test summary was cut off; that was wrong — see below.) The file is 66 MB and about 48 seconds long, so shrink it, and trim it toward 10 seconds if your instructor is strict about length.

You (Kabir)

[Uploaded Screen_recording_Before.mov and Screen_recording_after.mov]

upload these two to the github , under selab4

Claude

[Claude compressed both videos to 1080p (90 MB → 3.3 MB and 66 MB → 2.9 MB, full length, no cuts) and pushed them to SE_Lab4 as before.mov and after.mov in the commit "Lab 4: Add before and after videos".]

Correction: your after video does show the full test result. About 40 seconds in, "Ran 17 tests ... OK" is clearly visible; my earlier check only looked at the last frame.

Remaining: revoke the token, add the chat history PDF, and submit the Share link.

You (Kabir)

can you also make a file summarizing the changes made

Claude

[Claude wrote SE_Lab4/SUMMARY.md and pushed it in the commit "Lab 4: Add summary of changes".]

SUMMARY.md covers the five bugs and what caused each one, a table of all commits with hashes and changed files, what each task changed, a before-vs-after behaviour table, the design decisions, and how to run the game.

You (Kabir)

upload this too

Claude

No new file came through with your message. If you meant SUMMARY.md, it's already in SE_Lab4. If you meant the chat history PDF, attach it again.

You (Kabir)

okay can you give me this entire chat transcrpit i have to upload that also

Claude

[Claude produced this transcript as chat_history.pdf, with the GitHub tokens redacted, and uploaded it to SE_Lab4.]